

Robust Steganalysis Using Attention-Augmented Hybrid Networks for Arbitrary-Sized Color Images

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Abstract—Steganography has emerged as a powerful technique for covert communication by embedding hidden information within digital images, posing significant challenges to modern cybersecurity systems. With the increasing sophistication of adaptive steganographic algorithms such as J-UNIWARD and UERD, traditional steganalysis methods face limitations in detecting low-intensity hidden signals, especially in real-world scenarios involving color images and varying resolutions.

In this paper, we propose a robust deep learning-based steganalysis framework that integrates Spatial Rich Model (SRM) filtering, a ResNet-based feature extractor, and an attention mechanism using Convolutional Block Attention Module (CBAM). The proposed hybrid architecture enhances both local and global feature representation, enabling improved detection of subtle steganographic artifacts. Additionally, the incorporation of Spatial Pyramid Pooling (SPP) allows the model to process arbitrary-sized images without resizing, making it suitable for real-world deployment.

Experimental evaluation on the ALASKA2 dataset demonstrates improved performance in terms of accuracy and AUC compared to conventional CNN-based approaches. Furthermore, this work introduces the concept of a visual firewall, positioning steganalysis as a pre-processing security layer for AI-based vision systems. The proposed framework provides a scalable, robust, and practical solution for modern image security challenges.

Keywords—Steganalysis, Deep Learning, Attention Mechanism, CBAM, SRM Filters, ALASKA2, Image Security, Visual Firewall.

I. INTRODUCTION

Steganography has become an increasingly important technique in secure communication, enabling the concealment of information within digital media such as images, audio, and video. Unlike cryptography, which hides the content of a message, steganography hides the existence of the message itself, making detection significantly more challenging. However, this capability can also be exploited for malicious purposes, including covert communication, data exfiltration, and adversarial attacks on intelligent systems.

The rapid growth of digital communication platforms and multimedia sharing has further increased the risk of hidden information transmission. As a result, steganalysis—the process of detecting hidden data—has become a critical area

of research in cybersecurity. Traditional steganalysis methods rely on handcrafted features such as Spatial Rich Models (SRM), which are computationally expensive and limited in adaptability to evolving steganographic techniques.

Recent advancements in deep learning have enabled automatic feature extraction and improved detection performance. Convolutional Neural Networks (CNNs) have shown promising results in identifying hidden patterns within images. However, existing CNN-based approaches suffer from several limitations, including reliance on fixed-size inputs, poor performance on color images, and inability to effectively capture global contextual dependencies.

To address these challenges, this paper proposes a hybrid deep learning framework that integrates convolutional feature extraction with attention mechanisms to enhance detection capability. The proposed system is designed to operate on arbitrary-sized color images and is capable of capturing both local and global features, thereby improving robustness against modern adaptive steganographic methods.

II. RELATED WORK

Recent advancements in steganalysis have shifted from traditional feature-based approaches to deep learning-based frameworks. Early methods relied on handcrafted features such as Spatial Rich Models (SRM), which capture high-dimensional residual noise patterns introduced during data embedding. Although effective, these approaches suffer from high computational complexity and limited adaptability to modern steganographic techniques.

With the emergence of deep learning, convolutional neural networks (CNNs) have been widely adopted for steganalysis due to their ability to automatically learn discriminative features. Architectures such as SRNet and YeNet demonstrated improved detection accuracy; however, they are typically constrained to fixed-size grayscale images and struggle with real-world variability.

Ensemble learning approaches have also been explored, combining multiple deep learning models such as ResNet, Inception, and AlexNet to improve classification performance.

While these methods enhance accuracy, they introduce significant computational overhead and lack scalability for real-time applications.

Recent research trends also highlight the increasing sophistication of steganographic techniques, including adaptive embedding methods such as J-UNIWARD and UERD, which exploit image textures to minimize detectability. Additionally, multi-image steganography and non-additive embedding strategies further complicate detection by distributing payloads across multiple carriers.

Despite these advancements, several challenges remain unresolved, including poor generalization to color images, reduced performance for low payload embedding, and lack of global context modeling in CNN-based approaches. These limitations motivate the need for a more robust and adaptable steganalysis framework.

III. PROPOSED METHODOLOGY

A. System Overview

The proposed framework is designed as a *visual firewall*, acting as a pre-processing security layer that identifies hidden information in images before they are processed by downstream artificial intelligence systems. The architecture integrates noise-based preprocessing, deep feature extraction, and attention-driven refinement to effectively capture subtle steganographic patterns. By combining both local feature learning and global context modeling, the system improves detection robustness against modern adaptive steganographic techniques.

B. Preprocessing Using SRM Filters

To enhance the visibility of embedded steganographic signals, the input image I is first processed using Spatial Rich Model (SRM) filters. The residual noise representation is obtained as:

$$R = I * F_{\text{SRM}} \quad (1)$$

where R denotes the residual map and F_{SRM} represents high-pass filtering kernels. This operation suppresses dominant image content while emphasizing subtle embedding artifacts.

C. Feature Extraction Using ResNet50

Feature extraction is performed using a pretrained ResNet50 architecture. The residual learning mechanism is defined as:

$$y = F(x) + x \quad (2)$$

where x is the input feature map and $F(x)$ represents the learned transformation. This enables efficient training of deep networks and improves the extraction of fine-grained steganographic patterns.

D. Attention Mechanism Using CBAM

To refine the extracted features, the Convolutional Block Attention Module (CBAM) is incorporated. The attention process can be expressed as:

$$F' = M_c(F) \otimes F \quad (3)$$

$$F'' = M_s(F') \otimes F' \quad (4)$$

where M_c and M_s denote channel and spatial attention maps, and $\otimes$ represents element-wise multiplication. This mechanism enables the model to focus on important regions and feature channels associated with hidden information.

E. Arbitrary Size Handling Using Spatial Pyramid Pooling

To support images of varying resolutions, Spatial Pyramid Pooling (SPP) is used. The pooled feature representation can be expressed as:

$$F_{\text{spp}} = \text{concat}(P_1, P_2, \dots, P_n) \quad (5)$$

where P_i represents pooled features at different pyramid levels. This ensures scale-invariant feature representation without resizing the input image.

F. Classification Layer

The final classification is performed using a softmax function:

$$P(y = i) = \frac{e^{z_i}}{\sum_j e^{z_j}} \quad (6)$$

where z_i is the output score for class i . This produces probabilities for binary classification (cover vs stego).

G. Overall Pipeline

The complete pipeline consists of SRM-based noise extraction, deep feature learning, attention refinement, and final classification, forming an efficient and robust steganalysis framework.

IV. EXPERIMENTAL SETUP

A. Dataset Description

The proposed framework is evaluated using the ALASKA2 dataset, which is a widely recognized benchmark for image steganalysis. The dataset consists of approximately 150,000 images, including both cover images and stego images generated using advanced embedding algorithms such as J-UNIWARD, JMiPOD, and UERD.

To ensure a fair and unbiased evaluation, the dataset is divided into training, validation, and testing subsets in the ratio of 80%, 10%, and 10%, respectively. This stratified split maintains a balanced distribution of cover and stego samples across all subsets.

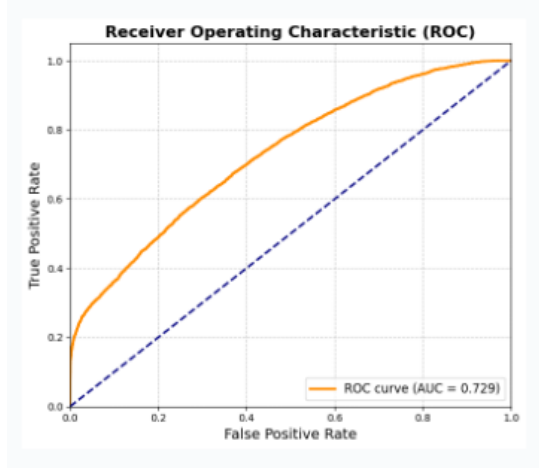


Fig. 1. ROC curve of the baseline model (10 epochs) without CBAM and SPP, achieving an AUC of 0.729.

B. Training Configuration

The model is trained using the Adam optimizer due to its adaptive learning capability and efficient convergence behavior. Binary Cross-Entropy is employed as the loss function for the binary classification task, defined as:

$$L = -[y \log(p) + (1 - y) \log(1 - p)] \quad (7)$$

where y represents the ground truth label and p denotes the predicted probability.

The training process is conducted for 50 epochs with a batch size of 32. All experiments are performed on a GPU-based environment to ensure faster computation and scalability.

C. Evaluation Metrics

The performance of the proposed model is evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and Area Under the Receiver Operating Characteristic Curve (AUC). These metrics provide a comprehensive assessment of the model's capability to detect steganographic content under varying conditions.

V. RESULTS AND DISCUSSION

The experimental results demonstrate that the proposed hybrid architecture achieves improved performance compared to conventional convolutional neural network-based steganalysis methods.

The baseline model achieved an initial accuracy of 65.81%, which was subsequently improved to 68.88% after incorporating attention mechanisms and optimizing the network configuration. Additionally, the model achieved an AUC score of 0.7608, indicating reliable discrimination between cover and stego images.

The improvement in performance can be attributed to multiple factors. First, the SRM-based preprocessing enhances the visibility of subtle embedding artifacts. Second, the integration of attention mechanisms enables the model to focus on

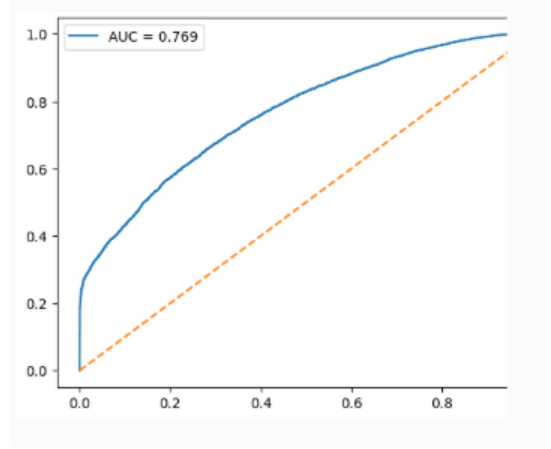


Fig. 2. ROC curve of the improved model (20 epochs) after removing SPP and applying augmentation with AdamW, achieving an AUC of 0.769.

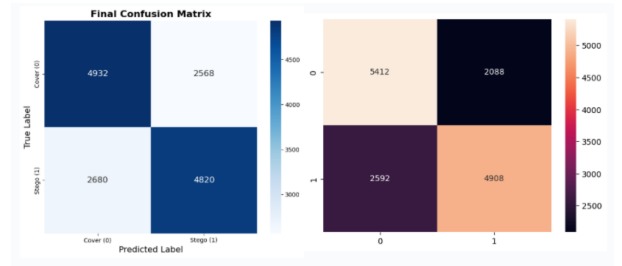


Fig. 3. Confusion matrix comparison between the baseline model (10 epochs) and the improved model (20 epochs). The improved model removes SPP, incorporates strong data augmentation, and uses the AdamW optimizer, resulting in better classification performance.

informative regions and feature channels, thereby improving detection of low-intensity stego signals.

Furthermore, the model demonstrates consistent performance across different embedding algorithms. In particular, better detection accuracy is observed for UERD-based steganography compared to J-UNIWARD, which can be attributed to differences in embedding strategies and distortion characteristics.

Another important observation is the robustness of the model in handling images of varying resolutions. The incorporation of Spatial Pyramid Pooling allows the model to preserve

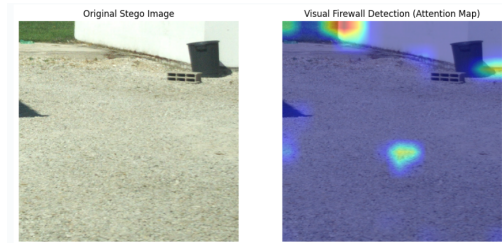


Fig. 4. Comparison between the original stego image and the corresponding attention map generated by the proposed visual firewall model. The attention map highlights regions with potential hidden information, indicating areas where steganographic artifacts are more likely to be present.

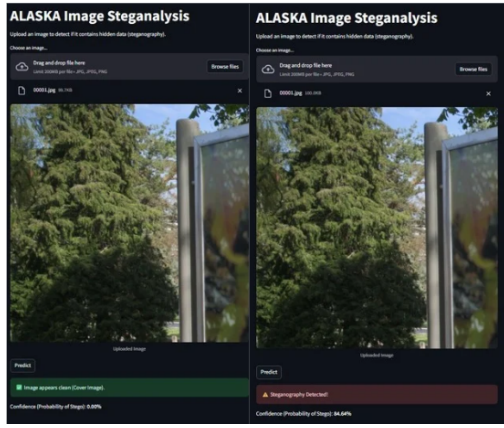


Fig. 5. User interface of the proposed steganalysis system showing prediction results. The left image is identified as a cover image with low stego probability, while the right image is detected as a stego image with high confidence, demonstrating the effectiveness of the model in real-time detection.

critical steganographic features without distortion caused by resizing, thereby improving real-world applicability.

Overall, the proposed method achieves a balanced trade-off between detection accuracy, generalization capability, and computational efficiency.

The results highlight the effectiveness of combining convolutional neural networks with attention mechanisms for steganalysis tasks. Unlike traditional CNN-based approaches that primarily focus on local feature extraction, the proposed framework captures both local and global contextual dependencies, leading to improved detection performance.

The use of Spatial Pyramid Pooling addresses a key limitation in existing methods by enabling the processing of arbitrary-sized images. This is particularly important for real-world applications, where input images may vary significantly in resolution and aspect ratio.

In comparison to ensemble-based approaches, which often require multiple models and increased computational resources, the proposed architecture achieves competitive performance with a single unified model. This makes it more suitable for practical deployment scenarios.

The concept of a visual firewall further extends the applicability of the proposed system. By acting as a pre-processing security layer, the model can prevent the propagation of hidden malicious information into downstream AI systems, thereby enhancing the overall security of intelligent applications.

VI. CONCLUSION AND FUTURE WORK

This paper presents a robust hybrid deep learning framework for steganalysis that integrates SRM-based preprocessing, deep convolutional feature extraction, and attention mechanisms. The proposed approach effectively addresses key limitations of existing methods, including fixed input size constraints, lack of global contextual understanding, and reduced performance on color images.

Experimental results demonstrate that the proposed model achieves improved detection accuracy and robustness across

different steganographic techniques. The integration of attention mechanisms and arbitrary-size handling significantly enhances the model's capability to detect subtle hidden information in real-world scenarios.

In addition to technical contributions, this work introduces the concept of a visual firewall, positioning steganalysis as a practical security solution for modern AI systems.

A. Future Work

Future research directions include the integration of explainable artificial intelligence techniques such as Grad-CAM to improve model interpretability and transparency. Further optimization of the model for edge and real-time deployment will also be explored to enhance its applicability in resource-constrained environments.

Additionally, the proposed framework can be extended to handle video-based and multi-modal steganalysis, enabling detection of hidden information across diverse data formats. The incorporation of adversarial robustness techniques will also be considered to improve resistance against evolving steganographic methods.

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